

GENERAL INTRODUCTION

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IN A NUTSHELL

- Intermittent rivers and ephemeral streams (IRES) are flowing waters that cease flow and/or dry at some point along their course
- Multiple natural and anthropogenic processes generate flow intermittence which influences almost every biogeochemical and ecological process in IRES
- IRES are dynamic mosaics of flowing, nonflowing water and terrestrial habitats, harboring distinctive aquatic, semi-aquatic, and terrestrial organisms that use different strategies to persist
- IRES suffer the same impacts of human activities as perennial streams and rivers but are often less valued
- A better understanding of the ecology of IRES will improve their legislative protection, restoration, and management

1.1 WHAT ARE INTERMITTENT RIVERS AND EPHEMERAL STREAMS (IRES)?

Flowing waters confined within a channel (except during floods) and moving in one direction are called rivers or streams. Rivers are considered to be larger and deeper than streams, although the distinction is a loose one of common usage rather than one based on fixed sizes and depths. The same common usage applies to describing differences in patterns of flow; “ephemeral” implies a shorter flow duration and lower predictability than “intermittent” but again there are no fixed boundaries. Thus, given the broad association of channel size and flow duration, a stream is more likely to be ephemeral and a river intermittent, prompting the distinction in the title of this book.

Of course, the association of channel size and flow is not this simple in reality (Chapter 2.1) nor is the classification of flow regimes (Chapter 2.2). The scientific literature is peppered with attempts (e.g., Uys and O’Keeffe, 1997; Williams, 2006; Gallart et al., 2012) to assign names to classes of streams and rivers whose flows cease for varying periods with varying predictability (Table 1.1). However, a global consensus remains elusive and probably will continue to do so, especially as many intermittently flowing waterways dry for widely different periods in different years, leading to variation among different categories within a single waterway. Therefore, rather than enter this semantic minefield, we refer to “intermittent rivers and ephemeral streams” and adopt the acronym “IRES” in this book as a shorthand term for all flowing waters that cease flow and/or dry completely at some point along their course.

Table 1.1 Examples of the diverse definitions of different types of IRES

Term	Definition	References
Ephemeral	Rivers that flow less than 20% of the time	Matthews (1988)
	Rivers that run for short periods after rain has fallen high in their catchments	Day (1990)
	Rivers that flow for less time than they are dry. Flow or flood for short periods of most years in a 5-year period, in response to unpredictable high rainfall events. Support a series of pools in parts of the channel	Uys and O’Keeffe (1997) Stringer and Perkins (2001)
	A channel formed by water during or immediately after precipitation events as indicated by an absence of forest litter and exposure of mineral soil	
	Rivers with dry stream beds in the dry season or even for longer periods	Bonada et al. (2007)
	Rivers/reaches receiving only river runoff because the channel is always above the water table	Larned et al. (2008)
	Rivers which form dry stream beds as the water disappears	Anna et al. (2009)
Episodic	Only filled after unpredictable rainfall and runoff. Surface water dries within days to weeks of filling and can support only short-lived aquatic life	Boulton et al. (2014)
	Streams and rivers—mostly in arid regions—which flow when unpredictable rain has fallen	Bayly and Williams (1973)
	Flow does not necessarily occur every year	King and Tharme (1993)
	Flow that only occurs after rainfall episodes	Davies et al. (1994)
	Highly flashy systems that flow or flood only in response to extreme rainfall events, usually high in their catchments. May not flow in a 5-year period or may flow only once in several years	Uys and O’Keeffe (1997) Williams (2006)
	Dry 9 years out of 10, with rare and very irregular flooding or wet periods which may last for a few months	
	Annual inflow is less than the minimum annual loss in 90% of years. Usually dry but filled after rare and large unpredictable rainfall events. Surface water persists for months to years and often supports longer-lived aquatic life (than ephemeral streams)	Boulton et al. (2014)
Temporary	Rivers with relatively regular, seasonally intermittent discharge	Boulton and Lake (1988)
	Rivers with intermittent discharge	Davies et al. (1994)
	Rivers in which the entire bed dries	Delucchi (1988)
	Flow stops and surface water may disappear along parts of the channel either yearly or during 2 or more years in 5	Uys and O’Keeffe (1997) Williams (2006)
	Natural bodies of water that experience a recurrent dry phase of varying duration	Acuña et al. (2014)
	Waterways that cease to flow at some points in space and time along their course	

Table 1.1 Examples of the diverse definitions of different types of IRES—cont'd		
Term	Definition	References
Intermittent	Streams or rivers which flow only seasonally and are otherwise dry	Bayly and Williams (1973)
	Rivers with intermittent discharge	Towns (1985) and Boulton and Lake (1988)
	Rivers that flow for 20%–80% of the year	Matthews (1988)
	Rivers that only dry in parts	Delucchi (1988)
	These rivers cease to flow and may dry along parts of their lengths for a variable period annually, or for 2 or more years in 5. Flow may recommence seasonally, or highly variably, depending on climatic influences and predictability of rainfall in the area. An intermittent river may experience several cycles of flow, no flow, and drying in a single year	Uys and O’Keeffe (1997)
	Rivers that hold water during wet portions of the year	Stringer and Perkins (2001)
	Natural bodies of water that experience a recurrent dry phase of varying duration	Williams (2006)
	Rivers with isolated pools during the dry season that may persist to the end of summer	Bonada et al. (2007)
	Rivers/reaches receiving groundwater when the water table intersects the channel and also might receive runoff	Larned et al. (2008)
	Rivers which form chains of lotic features (isolated pools) as flow disappears but surface water is still present	Anna et al. (2009)
Seasonal	Alternately wet and dry but less frequently or regularly than seasonal waters. Surface water persists for months to years and often supports longer lived aquatic life (than ephemeral streams)	Boulton et al. (2014)
	Flow that occurs for more than half the year, every year, during the same season	King and Tharme (1993)
Dryland	Alternately wet and dry every year, according to season. Usually fills and dries predictably and annually. Surface water persists for months, long enough for some plants and animals to complete the aquatic stages of their life cycles	Boulton et al. (2014)
	The rivers of arid and semiarid regions	Davies et al. (1994)
Interrupted	Rivers that flow for less than 20% of the time	Matthews (1988)
Nonperennial	Streams that dry out completely or for parts of their length during summer	Chester and Robson (2011)
Near permanent	Predictable flooding, though water levels may vary. The annual input of water is greater than the losses in 9 years out of 10	Williams (2006)
	Predictably filled, although water levels may vary. Annual inflow exceeds minimum annual loss in 90% of years. During extreme droughts, these waters may dry. Usually supports diverse aquatic life, much of which cannot tolerate desiccation	Boulton et al. (2014)
The main criteria used to separate them are duration and predictability of flow but these have not been used consistently among authors, leading to further confusion.		

We accept that by some definitions (Table 1.1), rivers can be ephemeral and streams intermittent, and that many authors prefer to refer to temporary or nonperennial streams and rivers. We also acknowledge there are many local names such as winterbournes, wadis, arroyos, and ramblas (e.g., Steward et al., 2012) used to describe IRES. This diversity of names is not surprising given the variety of IRES occurring worldwide (Fig. 1.1) and indicates that IRES are often recognized by local residents as unique parts of the landscape that deserve unique names.

Several other terms commonly used in this book to describe aspects of IRES also deserve early definition. The first is “flow cessation,” used to refer to the loss of surface flow along the channel, typically evident when riffles dry, resulting in chains of isolated pools (Fig. 1.2) that may or may not be connected by subsurface (hyporheic) flow. Periods of zero flow have important repercussions for biota and abiotic processes that rely on flow and that typify perennial streams and rivers, a key theme repeatedly discussed in the chapters to follow. The second term is “drying,” used to refer to the

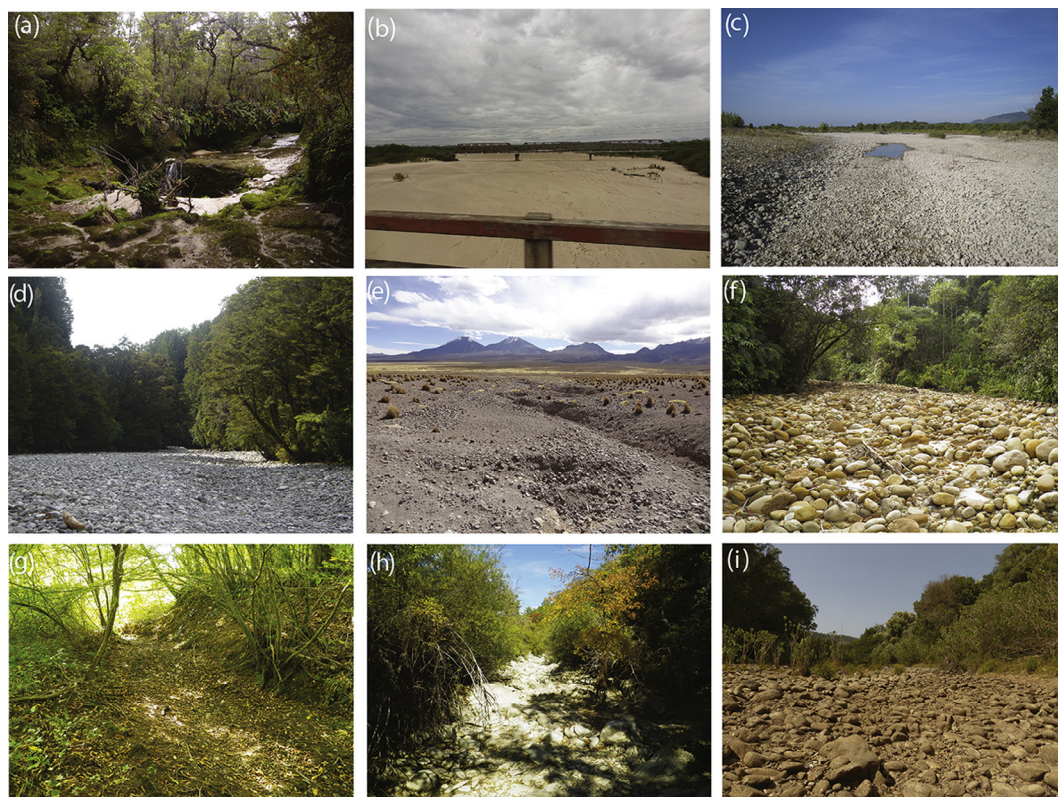


FIG. 1.1

Different types of IRES from across the world: (a) unnamed karstic stream, West Coast, South Island, New Zealand, (b) Río Seco, Chaco, Bolivia, (c) Asse River, Provence, France, (d) unnamed gravel-bed stream, West Coast, South Island, New Zealand, (e) unnamed stream, Altiplano, Bolivia, (f) Chaki Mayu, Amazonia, Bolivia, (g) Clauge, Jura, France, (h) Calavon, Provence, France, and (i) Río Hozgarganta, Andalucía, Spain.

Photos: Courtesy T. Datry (a–f), B. Launay (g and h), and N. Bonada (i).

**FIG. 1.2**

Four different hydrological phases in the Calavon River, southeastern France: (a) flowing, (b) nonflowing, (c) dry, and (d) flooding.

Photos: Courtesy B. Launay.

disappearance of surface water in the channel (Fig. 1.2), again with or without hyporheic flow. We prefer this term over “drought,” a word that is better reserved for the climatic process of drying and its broader effects (e.g., Lake, 2011) and over “drawdown” which is used in hydrogeology to refer to a decline in the water table. The third term frequently used in this book is “flow intermittence” (sometimes termed “intermittency”), referring to the temporal sequence of flow cessation and resumption that characterizes all IRES. In the chapters to follow, other specific terms may be used but will always include careful definition and explanation.

1.2 CAUSES OF FLOW INTERMITTENCE

Under natural conditions, many different processes generate flow intermittence in IRES (Table 1.2). As these processes often act in concert, disentangling their relative importance is difficult and probably unnecessary. However, some causes of intermittence predominate in different combinations of climate, water source, and channel features (e.g., width, bed permeability). For example, in alpine and arctic regions, flow ceases when water at the surface and in the shallow subsurface zone freezes (Robinson et al., 2016). IRES in the McMurdo Dry Valley in Antarctica (McKnight et al., 1999) flow for only a few months during austral summer before the onset of cooler weather freezes the free water and flow ceases. In contrast, flow intermittence in semiarid and arid regions is largely driven by low precipitation and high rates of direct evaporation and evapotranspiration through plants (Table 1.2). Flowing tracts of water in spatially intermittent IRES such as Sycamore Creek in the Sonoran Desert, Arizona, fluctuate widely in response to the temporal cycles of evapotranspiration and precipitation (Stanley et al., 1997).

In addition to climatic drivers, physical features of the channel, shallow alluvium, and catchment govern flow intermittence in many IRES (Chapter 2.1). Transmission loss is the infiltration of surface water into the streambed as a function of bed porosity and water table depth (Table 1.2). In IRES with coarse sediments (e.g., gravel-bed rivers) and water tables well below the channel (Sophocleous, 2002), transmission losses may be especially high, and these “losing” streams contribute substantially

Table 1.2 Types of IRES classified by predominant form of water loss

IRES type	Processes causing flow intermittence	Examples
Snowmelt and glacial melt water streams	Cessation of melting/ablation, freezing of stream surface and shallow subsurface water	Dry Valley streams, Antarctica; Val Roseg and Macun catchments, Switzerland
Perched and semiperched alluvial rivers	Transmission losses, depletion of bank storage, and floodplain aquifer	Albarine River, France; Selwyn River, New Zealand; Tagliamento River, Italy; Cooper Creek, Australia; Kuiseb River, Namibia; Mojave River, USA; Shashane and Wenlock Rivers, Zimbabwe
Nonperched rivers in arid and semiarid regions	Depletion of surface water and shallow groundwater by direct evaporation and evapotranspiration	Sycamore Creek, USA; Oued Chelif, Algeria
Low-order and headwater streams	Cessation of overland flow, depletion of saturated soil water or hillslope aquifer	Werribee and Lerderberg Rivers, Australia; Riera de Fuirosos, Spain; Maybeso Creek and Stillman Creek watershed, USA
Permafrost streams	Soil freezing, soil water, and wetland recession	Granger Creek and Wolf Creek, Canada
Karstic streams	Transmission loss, cessation of spring discharge, or groundwater inputs	River Lathkill and River Wye, UK; Coulazou River, France; Alme River, Germany
Lake outlets	Lake level drops below outlet elevation	Shadow Lake, Canada

Modified from Larned, S.T., Datry, T., Arscott, D.B., Tockner, K., 2010. Emerging concepts in temporary-river ecology. Freshwater Biology, 55, 717–738.

BOX 1.1 NONNATURAL CAUSES OF FLOW INTERMITTENCE

In addition to the many natural causes of intermittence, flow cessation can result from one or more human activities including alterations of land use, flow regulation, surface and/or groundwater extraction, and reduced precipitation and increased evaporation resulting from climate change (Palmer et al., 2008; Steward et al., 2012). Even large rivers are not immune to intermittence caused by humans. For example, the Colorado River, which flows 2334 km from its headwaters in the Rocky Mountains of Colorado to the Gulf of California, has been transformed from a perennial river to one that has flowed intermittently in most years since the completion of the Glen Canyon Dam in 1963 (Gleick, 2003). Another example of a large river becoming intermittent due to human activities is the Yellow River in China (Changming and Shifeng, 2002; Fu et al., 2004). This 5464-km river, which was perennial before 1972, is now intermittent and connects to the sea only infrequently. Flow intermittence is a direct result of the construction of 12 major dams along the river, coupled with a dramatic increase in water abstraction for direct human consumption and agricultural use. In addition to the barriers presented by the dams, longitudinal hydrological connectivity (Chapter 2.3) is also disrupted by an extended drying segment over 500 km long (Changming and Shifeng, 2002).

to aquifer recharge in many areas (Steward et al., 2012). Downward shifts in water tables in response to seasonal fluctuations and human activities can cause flow intermittence in streams or reaches that are normally “gaining” (Sophocleous, 2002) and may eventually lead to drying along the entire channel (Larned et al., 2008).

The upper reaches of most headwater and low-order streams flow intermittently, largely driven by rainfall (Benstead and Leigh, 2012). Channel flow ceases when overland flow stops and saturated soil water is depleted (Table 1.2); these processes are strongly influenced by vegetation cover, topographic relief, and the extent of bedrock in the upper catchment, along with rainfall. Streams draining lakes typically cease to flow when lake levels fall (Mielko and Woo, 2006) whereas IRES in karstic regions may flow intermittently in response to fluctuating groundwater discharge (Wood et al., 2005) (Table 1.2).

Intermittence also arises nonnaturally, caused by human activities (Box 1.1). In addition to converting perennial streams and rivers into IRES, these activities may impact on naturally IRES by lengthening the duration and extent of the zero-flow periods and altering their frequency and timing (Chapter 5.1). The interactions between natural and anthropogenic intermittence are poorly understood. This is especially true for the complex effects of altered flow regimes on sediment regimes (Chapter 2.1), water physicochemistry (Chapter 3.1), biogeochemical processes (Chapter 3.2), and the biota (Chapters 4.1–4.6) in IRES.

1.3 GLOBAL DISTRIBUTION AND AREAL IMPORTANCE OF IRES

IRES occur on all continents, including Antarctica. They are the predominant water bodies in hyper-arid, arid, semiarid, Mediterranean, and dry-subhumid regions, which together represent almost half of the Earth’s land surface (Tooth, 2000; Bonada and Resh, 2013). In many parts of the world, most of the channel lengths of natural waterways are intermittent. For example, over 70% of the 3.5 million km of river channels in Australia measured at the 1:250,000 scale are probably intermittent (Sheldon et al., 2010), and in American southwestern states such as Arizona and New Mexico, IRES comprise up to 94% of the total river length (Levick et al., 2008).

Even in temperate and humid areas, IRES are abundant. This is notably true for headwater streams. Headwater streams make up over 70% of the channel length of most river networks and are typically intermittent (Meyer and Wallace, 2001; Fritz et al., 2013; Datry et al., 2014), yet are often ignored when

regional estimates are made of flow regimes. For example, [Nadeau and Rains \(2007\)](#) used the National Hydrography Dataset to estimate that 59% of rivers and streams in the United States are intermittent. However, this dataset probably severely underestimated the total extent of IRES because it excluded stream segments less than one mile (1.6 km) in length and was based on 1:100,000 scale topographic maps ([Benstead and Leigh, 2012](#)). In addition to headwaters, many large IRES are reported from temperate and humid areas (e.g., [Buttle et al., 2012](#); [Snelder et al., 2013](#)). Indeed, IRES appear to be the most common flowing waters on Earth.

1.4 TRENDS IN A CONTEXT OF WATER SCARCITY AND CLIMATE CHANGE

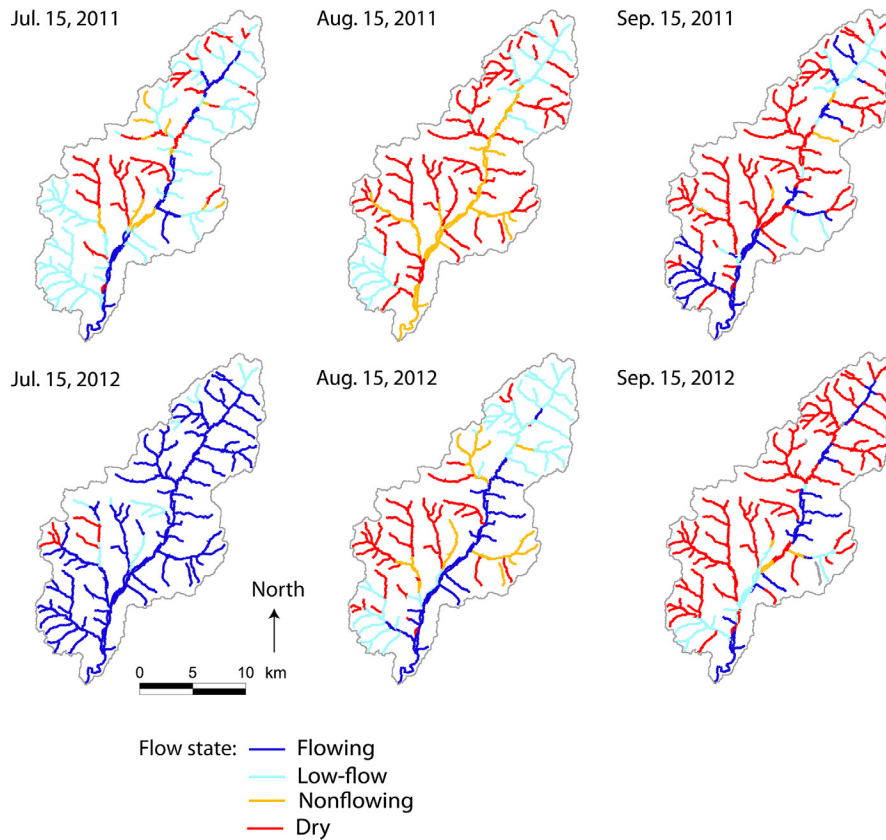
The flow regimes of streams and rivers are changing worldwide, mainly in response to changing climates coupled with rapidly increasing human demands for fresh water ([Chapter 5.1](#)). As a result, there is an ongoing global increase in the occurrence and spatial and temporal extent of IRES. This increase is likely to continue, with shifts from perennial to intermittent flow regimes projected by the 2050s for many parts of the world, including northeastern and southwestern Australia, Brazil, California, the Caribbean, southern Africa, West Africa, and around the Mediterranean Basin ([Döll and Schmied, 2012](#)). Conversely, some naturally IRES are predicted to become perennial, such as in Siberia and parts of Canada and Alaska due to warmer winters ([Döll and Schmied, 2012](#)).

Meanwhile, flows in many naturally IRES are declining and dry periods are getting longer. For example, in the intermittent Albarine River in temperate eastern France, hydrological modeling using various rainfall-runoff models coupled to climate change scenarios suggests that annual flow intermittence will increase on average (considering all Intergovernmental Panel on Climate Change scenarios) by 5% by 2050 ([Cipriani et al., 2014](#)). Furthermore, decreased annual precipitation (on average, 6%) and increased summer temperatures (on average, by 2°C) will extend dry periods both spatially and in duration.

The trend of increasing intermittence in IRES is not universal. Controlled releases from dams and weirs; discharge of agricultural, industrial, and urban effluents; and interbasin transfers have all contributed to decreased intermittence in many IRES, turning some of them perennial ([Hassan and Egozi, 2001](#); [Steward et al., 2012](#); [Datry et al., 2014](#)). In some water-scarce areas, the baseflow of many formerly intermittent urban rivers is now maintained by wastewater effluent ([Luthy et al., 2015](#)). In other areas, seasonal patterns of intermittence have been reversed. For example, where channels are used to carry irrigation flows released from upstream dams, peak flows from dam releases now occur when flow in the channel used to cease whereas instream dams now retain water when the downstream river historically flowed, sometimes causing flow to cease ([Barnett and Pierce, 2009](#)).

1.5 ECOLOGICAL FEATURES OF IRES

One distinctive feature of all IRES is that they are tightly coupled aquatic-terrestrial systems that are highly variable in both space and time as a result of flow intermittence and, in many cases, drying. In IRES over time, the wetting-drying cycle generates shifting habitat mosaics of lotic (flowing), lentic (nonflowing pools), and terrestrial (dry) habitats ([Datry et al., 2014](#); [Chapters 2.3 and 4.9](#)). All or any combination of these habitats can occur simultaneously and anywhere along the network ([Fig. 1.3](#)). As the habitats expand and contract, their spatial arrangement, temporal turnover, and hydrological connectivity constantly shift in response to changes in surface discharge and groundwater level ([Larned et al., 2010](#); [Datry et al., 2016a](#); [Fig. 1.3](#)).

**FIG. 1.3**

Example of a shifting mosaic of flowing, low-flow, nonflowing, and dry habitats in the Tude River, France.

From Datry, T., Bonada, N., Heino, J., 2016b. Towards understanding metacommunity organisation in highly dynamic systems. *Oikos*, 125, 149–159.

These habitats are colonized by different communities comprising taxa with preferences for each particular habitat. The regime of intermittent flow, wetting, and drying includes major disturbances for both aquatic and terrestrial organisms of IRES. Aquatic biota must contend with long periods of no flow or no surface water whereas when flow resumes, terrestrial biota must evade or tolerate inundation that may last for months or even years. Different strategies (and combinations of strategies) enable aquatic, semiaquatic, and terrestrial organisms to persist in IRES (Chapters 4.1–4.6). Some organisms withstand the disturbance *in situ* through different physiological or behavioral adaptations, collectively referred to as “resistance” (Chapter 4.8). Species from several aquatic groups can persist for months or years in dry river sediments as cysts, cocoons, or diapausing juveniles or adults (e.g., Boulton et al., 1992; Williams, 2006; Stubbington and Datry, 2013). Conversely, terrestrial organisms can have adaptations, such as the ability to float or to store air, that allow them to escape advancing wetting fronts or to resist submersion (Boumezzough and Musso, 1983; Adis and Junk, 2002; Lambeets et al., 2008).

Other organisms lack resistant stages but instead have the ability to quickly recolonize after disturbance, referred to as “resilience.” Examples include various life history traits and behavioral or morphological adaptations that promote high dispersal capacity, extensive use of refuges, short life cycles, and high fecundity (Fritz and Dodds, 2004; Williams, 2006; Datry et al., 2014). Some aquatic species with strong powers of dispersal quickly return to rewetted habitats by drifting from upstream waters or flying in from perennial refuges (Chester and Robson, 2011) whereas others recolonize from the hyporheic zone (Vander Vorste et al., 2016). Thus far, few strategies for resilience of terrestrial communities to inundation have been quantified in IRES, although several semiaquatic taxa avoid inundation by detecting rainfall cues (Lytle, 1999). It has also been suggested that some terrestrial organisms use advancing wetted fronts to colonize downstream terrestrial habitats (Corti and Datry, 2012; Rosado et al., 2014; Chapter 4.8).

Flow intermittence in IRES disrupts hydrological connectivity along longitudinal, lateral, and vertical dimensions (Chapter 2.3), creating a highly dynamic mosaic of habitat patches. These habitat patches are constantly being rearranged at multiple spatial and temporal scales, leading to perpetual fluctuations in the ecology and population or community structure of their residents. The temporal turnover and varying connectivity of these habitat patches in IRES render these ecosystems ideal arenas for studying metapopulations and metacommunities (Box 1.2) and disentangling the intricate ecological interactions among intermittence, biogeochemical processes, and biotic interactions.

BOX 1.2 IRES AS IDEAL ARENAS FOR METASYSTEM APPROACHES

Constant shifts in the spatial arrangement of habitat patches and their temporal turnover and connectivity generate recurrent ecological successions. These, in turn, create highly dynamic metapopulations and metacommunities (Jaeger et al., 2014; Cañedo-Argüelles et al., 2015; Datry et al., 2016a; Chapter 4.9). Recent work (Datry et al., 2016b) indicates that metapopulation and metacommunity patterns reported in perennial rivers are different in IRES, due to the effects of recurrent flow cessation events on population and community processes. For example, the role of regional (dispersal) processes may override that of local (environmental filtering) processes, especially following flow resumption when aquatic populations and communities are recovering (Chapter 4.9).

This could explain the general patterns of nestedness found in IRES. Nested patterns are those in which the species composition of less diverse assemblages (such as might be found in IRES) is a nested subset of more diverse assemblages (such as might be found in perennial streams and rivers). For example, several studies have shown that the number of fish species in IRES decreases as distance to perennial waters increases (e.g., Davey and Kelly, 2007; Colvin et al., 2009), and that many aquatic invertebrate communities are strongly nested along gradients of flow intermittence (Arscott et al., 2010; Datry et al., 2014). Conversely, local factors may predominate during initial phases of drying, and this could induce spatial turnover in species composition at the network scale (Chapter 4.9). Conceptual developments are needed to advance our understanding of metapopulations and metacommunities in dynamic systems such as IRES to guide effective management strategies.

Metasystem approaches are also highly relevant to understand how flow intermittence alters biogeochemical cycles. For example, organic matter dynamics in IRES are more pulsed than in perennial systems (Chapter 3.2); under lentic and dry conditions, the decomposition of leaf litter is strongly reduced or even halted (Corti et al., 2011). Large quantities of leaf litter accumulate on dry streambeds and are then entrained when flow resumes, along with myriads of terrestrial organisms (Corti and Datry, 2012; Rosado et al., 2014). The ecological consequences of these pulses are poorly understood but potentially range from generating anoxic “blackwater” events and altering downstream physicochemistry (Chapter 3.1) through to stimulating food webs downstream (Corti and Datry, 2012; Datry et al., 2014; Bixby et al., 2015; Chapter 4.7). Another major knowledge gap is understanding how the inclusion of IRES would alter current large-scale estimates of how much carbon is processed within river networks (Battin et al., 2009) or how much inland waters contribute to CO₂ outgassing (Raymond et al., 2013).

1.6 LEGISLATION, PROTECTION, RESTORATION, AND MANAGEMENT OF IRES

IRES face the same impacts from human activities (Fig. 1.4) as perennial streams and rivers. Broadly speaking, these impacts are alterations to the timing, volumes and durations of flow, changes to surface and groundwater quality, invasion by exotic species, and changes to the morphology of the channel and its catchment (Cooper et al., 2013; Acuña et al., 2014; Chapter 5.1), and all of them affect ecological processes and native biodiversity (Dudgeon et al., 2006). These effects interact in different ways and are superimposed on the projected effects of climate change to produce complex and often unpredictable alterations of sediment dynamics (Chapter 2.1), flow regime (Chapter 2.2), hydrological connectivity (Chapter 2.3), and water quality (Chapters 3.1 and 3.2). However, scientific research has focused far more on human impacts in perennial waterways than in IRES, and we have much yet to learn about their specific impacts in IRES, especially in the long term (Leigh et al., 2016).

The overwhelming focus on perennial streams and rivers may largely arise because they are more highly valued by the public than IRES (Chapter 5.2). For example, Armstrong et al. (2012), who interviewed landowners living along either perennial rivers or IRES in a small Pennsylvanian catchment, concluded that “Landowner perceptions and attitudes reveal a disproportionate lack of concern toward ephemeral or intermittent streams.” The ecosystem services of IRES, especially the more subtle ones of supporting and regulating services (Millennium Ecosystem Assessment, 2005), are currently little appreciated by society (Boulton, 2014), and therefore, it is not surprising that research on the likely effects of human activities has focused on perennial waterways.

Another reason for the focus on perennial waterways may be the difficulty of disentangling the ecological effects of natural flow intermittence from the impacts of particular human activities on aquatic biota. For example, the poor water quality associated with declining concentrations of dissolved oxygen that occurs naturally in drying pools (Chapter 3.1) resembles environmental conditions caused by nutrient enrichment from agricultural runoff or sewage entering a stream, and the effects upon resident aquatic biota are very similar. Standard biological monitoring techniques, using aquatic macroinvertebrates, for example, are seldom able to differentiate changes in lotic assemblage composition caused by organic pollution in IRES from the changes that occur naturally in response to flow cessation or drying (Sheldon, 2005). This has prompted investigations into ecological indices and tools that might be more powerful for assessing the ecological status of IRES (e.g., Mazor et al., 2014; Prat et al., 2014) and improve interpretation of different metrics (Cid et al., 2015) as well as approaches that incorporate IRES when defining reference conditions (Munné and Prat, 2011). Major challenges still exist, such as assessing when is the best time to sample or what metrics are the most reliable for detecting different human impacts on IRES (Chapter 5.1).

Ominously, a rather unfortunate assumption has often been drawn from the similarity of ecological responses by many IRES taxa to flow intermittence and to human impacts on water quality and flow. This assumption is that the biota of IRES are less likely to be adversely affected by human modifications of flow regime and water quality because the residents of IRES are more tolerant of “harsh” conditions and are more resilient or resistant to physical and chemical disturbances (Boulton et al., 2000). Although this may be true in some circumstances, it should not be used as an excuse for not assessing potential impacts of human activities on IRES when conducting environmental impact assessments or for assuming that IRES are somehow “second-class” waterways that can be ignored. The combination of ignorance of the ecosystem services and values of IRES and the assumed tolerance of their biota to

**FIG. 1.4**

Photos of different impacts in IRES in different parts of the world: (a) an IRES that is now perennial, carrying wastewater effluent (unnamed river, Fes, Morocco), (b) an IRES channelized for a golf course (unnamed river, Málaga, Spain), (c) an artificial IRES created by hydropower generation (Ter River, Llanars, Spain), (d) an urban IRES that is completely channelized (El Duí River, Collioure, France), (e) an IRES with an extensive olive tree plantation in the catchment (unnamed river, Valenzuela, Spain), (f) an IRES invaded by the exotic species *Arundo donax* (Barranc del Carraixet, Valencia, Spain), (g) an IRES heavily used by livestock (Barranc del Carraixet, Valencia, Spain), and (h) an IRES with urbanized river banks and used as a car trail during the dry period (Seco River, Vélez-Málaga, Spain).

Photos: Courtesy N. Bonada (a, b, d, e, h), P. Bonada (c), and N. Cid (f and g).

the impacts of human activities is probably the most serious threat currently jeopardizing the ecological condition of these waterways around the world (Acuña et al., 2014; Boulton, 2014).

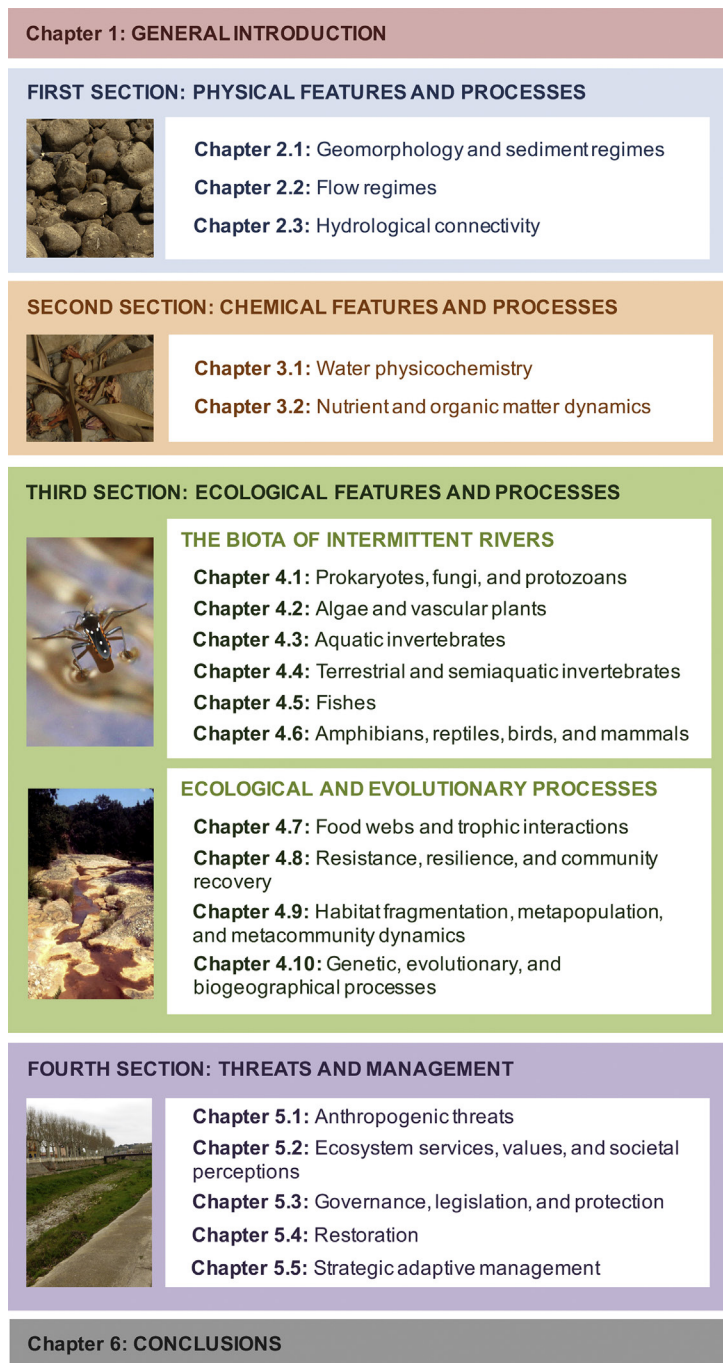
Far more research has been done on restoration and conservation of perennial streams and rivers than on intermittently flowing ones (Chapter 5.4). The few areas that have been protected, mostly in national parks, have usually been designed to conserve terrestrial landscapes and seldom adequately protect freshwater taxa (Abellán et al., 2007; Guareschi et al., 2012; Hermoso et al., 2015). In some countries, natural riverine reserves and freshwater protected areas are being implemented but rarely do they specifically target IRES. Furthermore, only a few countries have well-developed legislation that recognizes IRES, assisting their protection and wise management (Chapter 5.3). Finally, there have been very few efforts to specifically restore IRES damaged by human activities (Chapter 5.4), despite the widespread damage to these ecosystems (Leigh et al., 2016). In many cases, alterations of flow regime and water quality in IRES have been severe enough to produce such novel ecosystems that restoration back to some natural or near-natural condition is impossible and “reconciliation ecology” is seen as a more realistic framework for guiding their management (Moyle, 2014). Whatever guiding image is chosen, approaches such as strategic adaptive management (Chapter 5.5) are already improving the integrated management of IRES in some countries, building on the increased scientific understanding and legislative protection in recent years.

1.7 THE STRUCTURE OF THIS BOOK

The ongoing changes in flow patterns and global distribution of IRES raise a number of key questions: What are the effects of such changes on the ecology of IRES? How do these effects vary across different climates, flow durations, and landforms? How should we manage IRES to minimize the negative ecological effects of altered flow regimes yet optimize our use of their ecosystem services, especially access to potable fresh water? Are management strategies developed for perennial streams and rivers directly applicable to IRES? Or do refinements need to be made to acknowledge the centrality of flow intermittence to the ecology of IRES?

To answer these and other questions, we need to understand the physical, chemical, and biological features that influence the ecology of the biotic groups and ecosystem processes that characterize IRES, and the way that organisms in these systems have evolved. We also need to know the impacts that various human activities have upon IRES and the mechanisms (i.e., cause-effect relationships) behind these impacts. Thus, by using our expanding knowledge of the ecology of IRES, we can explore potential management approaches such as flow management, river restoration, and conservation. Using our understanding of the ecology of IRES to inform potential management approaches is a central theme of this book, and the chapters are arranged to follow this logical sequence (Fig. 1.5).

The first section, comprising three chapters (Fig. 1.5), describes the geomorphological and hydrological features of IRES. Chapter 2.1 outlines how different tectonic settings and lithology in different climates interact with the distinctive hydrology of IRES to generate a diverse array of sediment supply processes and channel geomorphology at multiple scales of space and time. The distinctive hydrological features, especially the intermittent and highly variable discharge, that characterize IRES and their flow regimes are discussed more fully in Chapter 2.2. In turn, the diversity of flow regimes in IRES creates spatial trends in longitudinal, lateral, and vertical hydrological connectivity that interact and vary over time (Chapter 2.3). These features and interactions influence almost every physical, chemical, and biological process occurring in IRES at local, catchment, and landscape scales.

**FIG. 1.5**

This book's structure showing the main sections and their constituent chapters.

Photos: N. Bonada (all except middle photo) and T. Herrera (middle photo).

The second section, comprising two chapters (Fig. 1.5), deals with the physicochemical and biogeochemical features of IRES. Spatiotemporal variations in flow regime and hydrological connectivity, particularly during low flows and drying, strongly influence water quality in IRES (Chapter 3.1). This is especially true of nutrient and organic matter dynamics at both reach and catchment scales, and there are also important links with hyporheic and riparian zone ecosystems as described in Chapter 3.2. The major influences of pulsed flows and, in many IRES, drying on physicochemistry and biogeochemical processes are a key theme in both chapters.

The third section (Fig. 1.5) describes the major groups of biota of IRES (six chapters) before reviewing the main ecological and evolutionary processes (four chapters). The ecology of the major groups is discussed primarily in terms of distribution, ecological roles, and interactions and commences with prokaryotes, fungi and protozoans (Chapter 4.1), and algae and vascular plants (Chapter 4.2). The major groups covered in these first two chapters play fundamental roles in organic matter production and decomposition in all IRES, with rates and fates being largely governed by water availability and flow intermittence. In turn, these major groups underpin the food webs and secondary production of IRES whose components include aquatic (Chapter 4.3) and semiaquatic and terrestrial invertebrates (Chapter 4.4); fishes (Chapter 4.5); and other vertebrates such as amphibians, reptiles, birds, and mammals (Chapter 4.6). Common themes throughout all these chapters include each group's ecological responses to rewetting and drying, the various modes of resistance and resilience, and the numerous adaptations to flow intermittence and variable habitat conditions.

Several of these ecological and evolutionary themes are explored in more detail in the next four chapters. Chapter 4.7 focuses on food web dynamics, trophic interactions, and how drying reduces the length of aquatic food chains but may initiate a switch to terrestrial energy pathways. Particularly important is the spatial and temporal fragmentation of habitats that substantially alters the strength of biological interactions such as predator-prey dynamics. This habitat fragmentation also necessitates various strategies for recolonization, resistance, and dispersal (Chapter 4.8) and creates a “dynamic mosaic” of lotic, lentic, and terrestrial habitats in IRES, providing opportunities to test theories about metapopulations, metacommunities, and metaecosystems (Chapter 4.9, Box 1.2). These three levels of system organization are subjected to different evolutionary pressures in IRES resulting in the various species traits and trade-offs described in Chapter 4.10.

The fourth and final section of the book describes the challenges of environmental management in IRES and complements the preceding sections on physical, chemical, and ecological aspects. It comprises five chapters (Fig. 1.5). The many human activities that impact upon IRES are discussed in Chapter 5.1, with particular emphasis on alterations of flow regime, water quality, biota, and ecological processes. Appreciating what is being lost through lack of appropriate management involves understanding and evaluating the various ecosystem services provided by IRES to help address societal perceptions that currently undervalue these assets (Chapter 5.2). Chapter 5.3 describes some of the current legislative frameworks that, in many parts of the world, still fail to fully acknowledge the importance of IRES and their significance to river networks. However, attitudes and legal perspectives are changing, and this chapter also reviews the integration of ecological insights and systems thinking with jurisdictional changes in effective water governance to better protect threatened assets. The fourth chapter in this section explores the restoration of IRES, drawing contrasts, and parallels with the restoration of perennial streams and rivers (Chapter 5.4). Drawing together these threads of mitigating human impacts on IRES against a backdrop of societal ignorance of these ecosystems' values, the legislative limitations in many countries, and the limited understanding of restoration approaches, Chapter 5.5

outlines strategic adaptive management as a way to link science to value-driven objectives. In this way, management strategies both use and inform the physical, chemical, and biological insights presented in the first three sections, and the final chapter of this section highlights the benefits of successful integration of science and management of IRES and their surrounding landscapes.

Principal themes and conclusions from all of the chapters in these four sections are summarized in the book's last chapter (Fig. 1.5). It is important to realize that this research field is rapidly growing, new insights are published frequently, and inevitably some of the new information will challenge ideas and perceptions presented in this book. This is to be expected and is a healthy sign of any fertile discipline. Nonetheless, a global synthesis like this one is crucial to take stock of what is known, collate the currently scattered information on IRES from diverse parts of the world, and provide the first book solely devoted to describing the ecology and management of one of Earth's most common aquatic ecosystems.

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